FABIEN SANGLARD'S WEBSITE

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TO BECOME A GOOD C PROGRAMMER

Every once in a while I receive an email from a fellow programmer asking me what language I used for one of my games and how I learned it. Here is an entry that list the best things to read about C.

If you know of other gems, please email me or add a comment at the bottom of the page.

The answer (you can skip this)

As I mentioned it in a previous entry, all the commercial 3D engines I wrote so far are 95% C89 (aka "Standard C", aka "ANSI C"). I picked C89 instead of C99 because some compilers still don't support fully C99…and to be able to compile on iOs, Windows and Xbox 360 was a mandatory from day one.

Depending on the compiling platform, the 5% remaining are Objective-C (iOS) or C++ (Windows, Mac OS X) to bind the engine to the native inputs/output. Unexpectedly the C vs C++ proved quite controversial on reddit. The two real reasons were:

- I didn't know C++ well enough at the time I started writing the engines. The platform was quite limited (iPhone 2G) and I knew I would need to totally master my tools in order to achieve 60 frames per second.
- Since I first read id Software source code, it was a dream to complete a commercial game using C.

Were those "good" choices? I think in the end the only valid questions are "Did you ship it?" and "Was it fast?". Considering the incredible frame rate achieved (some people mentioned experimenting motion sickness while playing Shmup) I think I made the right calls.

Bad C readings (stop skipping)

I'm going to start with the things I didn't take too seriously: Internet tutorials, blogs and almost anything brought by Google (yes, it includes this article). I usually considered those sources unreliable and potentially harmful.

Like a lot of people in the industry I used to Google way too often. Overtime I found the illusion of speed and the inaccuracy of the answers to be counter-productive.

No website is as good as a good book. And no good book is as good as a disassembly output.

Excellent C readings

"C Programming Language" (aka: K&R)

The classic and first book you should read about C.

It will be easy to pick up as it is a 272 pages, 386 grams book. Short and full of well explained code samples it was written by the fathers of the language Kernighan and Ritchie themselves. It is all you need to know about C…for the first few weeks. It is fun to read, keeps things short and will get you going in no time. You will probably skip the Annex A (about obscure things like promotions, decaying, conversions and other useless things) and Annex B about the C Library…and I

think it's ok for the beginning. This book makes C appears very small and simple so it is very encouraging to learn.

If you keep on practicing and learning you will soon hit some strange situations. Like for example:

```
unsigned int   ui_one       =  1 ;
signed   int   i_one        =  1 ;
signed   short s_minus_one  = -1 ;

if( s_minus_one > ui_one)
            printf("-1 > 1 \n");

if( s_minus_one < i_one)
            printf("-1 < 1 \n");

#./run
#
# -1 > 1
# -1 < 1
```

In the previous code sample, due to "integral promotion" -1 was once evaluated to be greater than 1 and then smaller than 1. C has plenty of case like this when the language cannot longer "do stuff" for you.

There are also plenty of subtleties:

```
extern void foo(void);
void (*f)();

f = &foo;   // Valid
f = foo;    // Valid too ! (syntactic sugar)

f();        // Calls f
(*f)();     // Calls f too ! (syntactic sugar)
```

Or the array/pointer/decaying special cases...

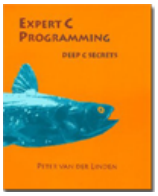
```
int array[] = {0,1,2,3,4} ;
int* pointer = array;

if (sizeof array == sizeof pointer)
        printf("This will never be printed !!");

if (sizeof(int*) == sizeof &array[0])
        printf("This will be printed !!\n");

if (&array[2] - &array[0] == 8)
        printf("This will never be printed either, result is 2 not 8!!");
```

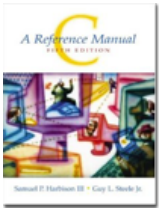
When you hit the point where you understand that you actually know very little about C (and that the Annex A is way too slim), it will be time to pickup the second book.



Expert C Programming

This book is fantastic because it will bring your attention to what happens under the hood in a very entertaining way. Through numerous bug anecdotes and trivia (mostly NASA based) the reader will be introduced again to integral promotion, subscripting, decaying and many other C marvels. This book is so captivating that you will probably read the 353 pages within a night and be dissapointed it was so short.

Now willing to do deep and become a good C programmer, you should acquire the last book you will ever need:

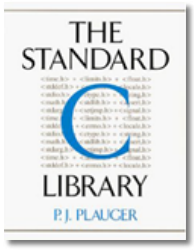


C: A Reference Manual

This is the ultimate C/C89/C99 book. The true cold boring truth that you will deal with from now on. You can put K&R and ECP back on the shelves and keep this one besides the screen monitor. Anything you want to know is there.

The C Standard library

February 6th 2013 : Two years have passed since I wrote this article. The three books mentioned remain the best in my opinion ... but what about the C Standard Library ?



The best book to master the C Library is with no doubt "The Standard C Library" by P.J. Plauger : Not only it comes with an implementation of the entire library, it also discuss design decisions and provides historical perspectives (ever wondered why C performed all floating operations in `double` ? Or how `errno` came to existence ? This book has the answers.

It covers in details the 15 modules of C89: `assert.h`, `ctype.h`, `errno.h`, `float.h`, `limits.h`, `locale.h`, `setjmp.h`, `signal.h`, `stdarg.h`, `stddef.h`, `time.h`, and discuss at length the "big four": `stdio.h`, `stdlib.h`, `string.h` and `math.h`

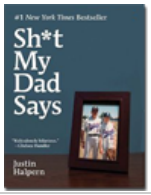
.

The only two big things missings are the coverage of C99 integral types (`intypes.h` and `stdint.h`) which are paramount in order to write portablecode and C11's `threads.h` which is equaly important considering the emergence of multi-core CPUs. For now I haven't found anything better but I welcome reader's suggestions.

Additional readings

id Software codebase

Of course reading books is not enough. To read great C code will help tremendously. My favourite is id Software's 3D engines codebase: Doom, Quake, Quake 2, Quake 3, Wolfenstein 3D iPhone and Doom iPhone. While I read them I wrote memos for myself which I later turned into articles (Doom, Quake,Wolfenstein 3D iPhone and Doom iPhone).



Sh*t My Dad Says

Try to maintain an healthy life and read some fun stuff for a change ;) !

Comments

39 Comments **Fabien Sanglard's website** **Sohaib Mohammed**

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sylvainulg • 8 years ago

I'd tend to recommend -Wall as an additional reading. Enabling sufficient warnings in C is very important to have reminders of pitfalls, such as "this loop will never end because you're comparing a unsigned to -1, and that's always true", or "I doubt you want to assign a new value to X here, and more likely want to compare X to that value. Use if ((x=foo(bar)) ... if that really was your intent".

8 • Reply • Share ›

Wondersonic • 8 years ago

Another great post, thanks!!!

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Ben • 8 years ago

Thx, this will help me a lot learning C :) I'm just diving into it.

• Reply • Share ›

Antonio • 8 years ago

Hi, i'm reading the c programming language book by Kerninghan and Ritchie, you said that this book is easy, i don't think so, especially if you try to do the exercises using the techniques explained.
Yes there are chapters that are easy and others don't, like the last example of the pointers chapter and the exercise 5-20.
It is only 200 pages and is written by the author of C, this is why i like this book.
I think this is the only book a C programmer need, after it will be a good idea read the alghoritms in C book by Sedgewick Robert. and when you start to build some project code complete by Steve McConnell.

4 • Reply • Share ›

Albert1 • 8 years ago

It seems you like to use OOP principles in C... I think you should read "Object Oriented Programming with ANSI C" by Axel-Tobias Schreiner, now freely downloadable at http://www.cs.rit.edu/~ats/...

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Albert1 • 8 years ago

Another good book is "C Interfaces and Implementations: Techniques for Creating Reusable Software" by David R. Hanson.

1 • Reply • Share ›

RenÃ© Hansen • 7 years ago

Good read and thanks for the book recommendations. Only have K&R at the moment, will check out the other two.

• Reply • Share ›

anon • 7 years ago

I recommend two other books:

1. Steve Maguire's Writing Solid Code (good practices for reliable C programming)
2. P.J. Plauger's The C Standard Library (implementation of clib without the platform cruft you will see on your home machine sources)

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noop • 7 years ago

Hi. I'm one of those C++ programmers who use C++ as a convenient extension to C.It's really good, if used carefully. After reading discussion on reddit I agree with GoAwayStupidAI user, C++ can be used to generate code at least as efficient as C code in less time. Fact that C++ lets you do a lot of really horrible stuff doesn't mean that you can't constrain youself to a subset that really works.

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DaveK • 6 years ago

I beg to differ with you about the third paragraph in your second example. Since f is a pointer (to a function), the standard way of accessing the object it points to is by dereferencing it with an asterisk. So I'd call (*f)(); the standard way, and the ability to invoke it directly without dereference operator by using just plain f(); is the syntactic sugar.

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Asif Malik • 6 years ago

Thanx a lot for such a wonderful article... I will try to manage these books....

• Reply • Share ›

Tacey Wong • 6 years ago

English-speaking people has a great advantage in programing.(æ¥è‡ªä¸å›½ç„ä¸€åå-¦ç"Ÿå'æ,¨é—®å€™ï¼‰

3 • Reply • Share ›

Shay • 6 years ago

Your function pointer example left out the most fun parts (comments converted to C89 style so it would compile with gcc -std=c89 -pedantic):

so it would compile with gcc -std=c89 -pedantic).

```
extern void foo(void);
void (*f)();

f = &foo /* Valid */
f = foo; /* Valid too ! (syntactic sugar) */

f(); /* Calls f */
(*f)(); /* Calls f too ! (syntactic sugar) */

foo(); /* usual way of calling function itself */
(*foo)(); /* you like verbosity, huh? */
(**foo)(); /* wha? */
(****************foo)(); /* well, um, OK */
(**&****&***&**&*&****&*foo)() /* just don't put two & together */

(**&***&*f)() /* same for function pointer */
```

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Shay • 6 years ago

Sorry to double-post, but another C book I highly enjoyed was Andrew Koenig's "C Traps and Pitfalls". Lots of things to keep you alert while coding in C.

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juhi • 5 years ago

still confuse which book should be prefer for C plz help
i want be a good programer & i work really hard for it still i think i m not satisfy plz guide me

• Reply • Share ›

Kirill → juhi • 5 years ago

are you sure you wanna be c programmer?

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Anaël • 5 years ago

The only thing that make me missing c++ is operator overloading for vector and matrix manipulation, it just make code easier to read. What is your approach about that ? Is there an elegant approach in c ? Is it so bad to use c++ as a "c with operator overloading" ?

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Aysha • 5 years ago

One must remember that learning any language in terms of programming its often quite tough task to execute the whole process in timely manner as well as knowing the entire language part how to implement in the process of making the program perfect this is obviously needed for all. As per above guidelines learning c is basically the hard liner but this processes are usually good one to grove and learn the technique nicely plus smoothly. I pay someone to do my paper on c program recently. One could draw the conclusion that the entire course of actions is very much important to learn anything new. Thanks for sharing.

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Oscar J • 4 years ago

Great post!. The C Programming Language is indeed a great book to have. There is also my favorite called A modern Approach that is amazing. I have gathered my favorite resources on the subject here:
Best C Programming Books

Feel free to check it out.
Great post again, Thx :)

1 • Reply • Share ›

Misu Popescu • 4 years ago

Hello,
just wanted to know what is your opinion of "Learn C the Hard Way" by Z. Shaw (the book is also published for free online by the author) as a beginner C book.
Thanks

• Reply • Share ›

Vivek Pandey • 4 years ago
which is the best book to understand c language
Reply • Share ›

Enrico Passoni • 4 years ago
great post
Reply • Share ›

Nabillan • 4 years ago
i start to learning C programming,but sometimes i can't understand the code,can u help me?
Reply • Share ›

Zulu Lala • 4 years ago
Actually one of the best reads I've had to help learn C and programming in general was a book about assembly! You can find it here: http://www.drpaulcarter.com...

Going so low really helps to understand the basics like signed/unsigned comparisons, bit operations, pointers, why the size of int can change, etc.
1 Reply • Share ›

sankar prasad biswas • 3 years ago
new here. want a c code that can evaluate a infix expression like (2.5+2) floating point evaluation is tough for me. (please use infix to post fix then evaluate and using array implimentetion.)
Reply • Share ›

irishguyny • 3 years ago
you could always check out the free programming ebooks at http://webwork.io/ebooks
Reply • Share ›

wtf2020lol • 3 years ago
Always do what makes you happy! Programming opinions are like song requests told to a DJ, it's impossible to make everyone happy, so ignore the idiots and assholes, then play the songs that seems right to you.
Reply • Share ›

weishigoname • 3 years ago
nice post, thanks for sharing
Reply • Share ›

Devora Decker2 • 3 years ago
Savvy writing , I learned a lot from the facts ! Does someone know where I can acquire a template USCIS I-9 version to work with ?
Reply • Share ›

help with physics • 3 years ago
The ideas and tricks that you give on how to program using C programming language is really useful and effective that it can give someone some proper information that can surely guide them a lot in the end. Through this, it would be easy for them to become a good programmer on it.
Reply • Share ›

behemoth • 3 years ago
A great book is "Pointers on C" by Kenneth Reek.
Reply • Share ›

Abhijit Paul • 3 years ago
What is the better IDE / tool for writing Programming in C in Windows environment
Reply • Share ›

Jibril Hartri Putra • 2 years ago
thank you very much :) it helps my lab (http://ilab.gunadarma.ac.id)
Reply • Share ›

marco • 2 years ago

marcc • 2 years ago

The book on C that has helped me the most is Allen I. Holub's The C Companion (Prentice-Hall 1987), in particular all the technical background information provided. For example, the starting chapter on binary arithmetic is invaluable.

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Dean Hammond ➦ marcc • 4 months ago

Totaly agree - his "take" on the subject is thorough and the chapter on pointers and more complex declarations is a life saver. A worthy addition to K&R and Co.

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let's learn • 2 years ago

It is told that c programming is the mother of all languages. c programming is used in most of the important sectors. All the operators are made by c programming . So, it is very important to learn c programming .
In my website c programming is taught in very easy way. You will get videos too in my website ltslearn.com.

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Romain • a year ago

Hello, first thanks.
Second, have you read "21st Century C" ?
It has a lot of good comments, what do you think about it ?

• Reply • Share ›

rebecca zhu • 10 months ago

C11 is out and C99 isn't even fully supported... What has this world come to?

Fabien Sanglard @2011